

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Linear Algebra (Vectors & Independence), K-Means Clustering, Transactions & ACID, Data Interpretation

Contents

1	Topic 1: Vectors, Vector Spaces & Linear Independence	2
1.1	10 Previous Year Questions (PYQs)	2
1.2	10 Practice Questions (GATE Level)	4
2	Topic 2: Clustering: K-Means (Math + Lloyd's Algorithm)	7
2.1	10 Previous Year Questions (PYQs)	7
2.2	10 Practice Questions (GATE Level)	9
3	Topic 3: Transactions & ACID Properties	11
3.1	10 Previous Year Questions (PYQs)	11
3.2	10 Practice Questions (GATE Level)	13
4	Topic 4: Data Interpretation – Tables & Bar Graphs	16
4.1	10 Previous Year Questions (PYQs)	16
4.2	10 Practice Questions (GATE Level)	19

1 Topic 1: Vectors, Vector Spaces & Linear Independence

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Consider the following set of vectors in $\mathbb{R}^3$:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 7 \\ c \end{bmatrix}$$

For what real value of c is the set $\{v_1, v_2, v_3\}$ linearly dependent? **Answer: 10** **Solution:** Vectors are linearly dependent if and only if the determinant of the matrix formed by them as columns is zero:

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & c \end{bmatrix} = 0$$

Expanding along Row 1: $1(5c - 49) - 2(2c - 21) + 3(14 - 15) = 0 \implies (5c - 49) - (4c - 42) + 3(-1) = 0 \implies c - 7 - 3 = 0 \implies c = 10$.

Q1.2

[GATE DA 2024 | MCQ | 1 Mark]

Let V be a vector space of dimension n over $\mathbb{R}$. Which of the following statements is FALSE?

- (A) Any subset of V containing more than n vectors is linearly dependent.
- (B) Any linearly independent subset of V containing exactly n vectors spans V .
- (C) Any spanning set of V containing exactly n vectors is linearly independent.
- (D) Any subset of V containing fewer than n vectors cannot be linearly independent.

Answer: (D) **Solution:** A subset containing fewer than n non-zero vectors can certainly be linearly independent (e.g., a single non-zero vector in $\mathbb{R}^3$ is linearly independent). However, it cannot span an n -dimensional space. Thus, statement (D) is false.

Q1.3

[GATE CS 2023 | NAT | 2 Marks]

Let $S = \{x \in \mathbb{R}^4 \mid x_1 + 2x_2 - x_3 + 3x_4 = 0 \text{ and } 2x_1 + 4x_2 - 2x_3 + 6x_4 = 0\}$. What is the dimension of the subspace S ? **Answer: 3** **Solution:** Notice that the second equation $2(x_1 + 2x_2 - x_3 + 3x_4) = 0$ is a scalar multiple of the first equation. Thus, there is only 1 linearly independent linear constraint on $\mathbb{R}^4$. By the Rank-Nullity theorem, $\dim(S) = n - \text{rank}(A) = 4 - 1 = 3$.

Q1.4

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following subsets of $\mathbb{R}^3$ is/are subspaces of $\mathbb{R}^3$?

- (A) $W_1 = \{(x, y, z)^T \in \mathbb{R}^3 \mid 2x - 3y + z = 0\}$
- (B) $W_2 = \{(x, y, z)^T \in \mathbb{R}^3 \mid x + y + z = 1\}$
- (C) $W_3 = \{(x, y, z)^T \in \mathbb{R}^3 \mid xy = 0\}$
- (D) $W_4 = \{(x, y, z)^T \in \mathbb{R}^3 \mid x = 2y \text{ and } y = 3z\}$

Answer: (A, D) **Solution:** (A) represents a plane passing through the origin (closed under vector addition and scalar multiplication). (B) does not contain the zero vector $(0, 0, 0)^T$. (C) is the union of coordinate planes and is not closed under addition: e.g., $(1, 0, 0) + (0, 1, 0) = (1, 1, 0)$, where $xy = 1 \neq 0$. (D) is a line passing through the origin (the intersection of two hyperplanes passing through origin), which is a valid 1-dimensional subspace.

Q1.5

[GATE CS 2022 | MCQ | 1 Mark]

Let u, v, w be three linearly independent vectors in a vector space V . Consider the set $S = \{u + v, v + w, w + u\}$. The set S is:

- (A) Always linearly independent over $\mathbb{R}$.
- (B) Always linearly dependent over $\mathbb{R}$.
- (C) Linearly independent only if u, v, w are mutually orthogonal.
- (D) Dependent if the dimension of V is 3.

Answer: (A) **Solution:** Set $c_1(u+v) + c_2(v+w) + c_3(w+u) = 0 \implies (c_1+c_3)u + (c_1+c_2)v + (c_2+c_3)w = 0$. Since u, v, w are independent, $c_1 + c_3 = 0, c_1 + c_2 = 0, c_2 + c_3 = 0$. The determinant of this system is $\det \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = 1(1) + 1(1) = 2 \neq 0$. The only solution is $c_1 = c_2 = c_3 = 0$, so S is always linearly independent.

Q1.6

[GATE DA 2025 | NAT | 2 Marks]

Let A be a 4×3 real matrix with rank 2. What is the dimension of the null space (nullity) of A^T ? **Answer: 2** **Solution:** Matrix A^T has size 3×4 . The rank of A^T equals $\text{rank}(A) = 2$. By the Rank-Nullity Theorem applied to A^T : $\text{rank}(A^T) + \text{nullity}(A^T) = \text{number of columns of } A^T = 4$. Therefore, $\text{nullity}(A^T) = 4 - 2 = 2$.

Q1.7

[GATE CS 2021 | MCQ | 2 Marks]

Let P_2 denote the vector space of all polynomials of degree at most 2 over $\mathbb{R}$. Consider the polynomials $p_1(t) = 1 + t, p_2(t) = t + t^2, p_3(t) = 1 + t^2$. Which of the following statements is TRUE?

- (A) $\{p_1, p_2, p_3\}$ forms a basis for P_2 .
- (B) $\{p_1, p_2, p_3\}$ is linearly dependent.
- (C) $p_1 + p_2 = p_3$.
- (D) The dimension of $\text{span}(\{p_1, p_2, p_3\})$ is 2.

Answer: (A) **Solution:** Coordinate vectors with respect to standard basis $\{1, t, t^2\}$ are $[1, 1, 0]^T, [0, 1, 1]^T$, and $[1, 0, 1]^T$. The determinant is $\det \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = 2 \neq 0$. Since they are 3 linearly independent vectors in a 3-dimensional space P_2 , they form a valid basis.

Q1.8

[GATE DA 2026 | NAT | 1 Mark]

The dot product of two vectors $u = [1, 2, -1]^T$ and $v = [2, k, 4]^T$ is zero. What is the value of k ? **Answer: 1** **Solution:** $u \cdot v = (1)(2) + (2)(k) + (-1)(4) = 2 + 2k - 4 = 2k - 2 = 0 \implies 2k = 2 \implies k = 1$.

Q1.9

[GATE CS 2020 | MSQ | 2 Marks]

Let A be an $m \times n$ matrix with real entries. Which of the following statements is/are ALWAYS true?

- (A) The row space of A and the null space of A are orthogonal complements in $\mathbb{R}^n$.
- (B) $\dim(\text{Col}(A)) = \dim(\text{Row}(A))$.
- (C) If $m < n$, the columns of A must be linearly dependent.
- (D) If $m > n$, the rows of A can be linearly independent.

Answer: (A, B, C) **Solution:** (A) Fundamental theorem of linear algebra: $\text{Row}(A) \oplus \text{Null}(A) = \mathbb{R}^n$. (B) Row rank equals column rank. (C) There are n column vectors in $\mathbb{R}^m$; since $n > m$, any set of $> m$ vectors in $\mathbb{R}^m$ is linearly dependent. (D) is false because there are m rows in $\mathbb{R}^n$, and since $m > n$, the rows must be linearly dependent.

Q1.10**[GATE DA 2024 | NAT | 2 Marks]**

What is the dimension of the subspace of $\mathbb{R}^{3 \times 3}$ consisting of all 3×3 real skew-symmetric matrices ($A^T = -A$)? **Answer: 3** **Solution:** A skew-symmetric matrix has all diagonal entries equal to zero ($a_{ii} = -a_{ii} \implies a_{ii} = 0$) and entries below the diagonal strictly determined by entries above: $a_{ji} = -a_{ij}$. The number of free independent entries is the number of strictly upper triangular entries: $\frac{n(n-1)}{2} = \frac{3(2)}{2} = 3$.

1.2 10 Practice Questions (GATE Level)**P1.1****[Practice | MCQ | 1 Mark]**

Let v_1, v_2, v_3 be vectors in $\mathbb{R}^4$. If $2v_1 - 3v_2 + v_3 = 0$, then:

- (A) $\{v_1, v_2, v_3\}$ is linearly dependent.
- (B) $\{v_1, v_2, v_3\}$ is linearly independent.
- (C) The vectors must all be zero vectors.
- (D) $\text{span}(\{v_1, v_2, v_3\}) = \mathbb{R}^4$.

Answer: (A) **Solution:** A linear combination with non-zero scalar coefficients ($c_1 = 2, c_2 = -3, c_3 = 1$) equals zero, which directly satisfies the definition of linear dependence.

P1.2**[Practice | NAT | 2 Marks]**

Let $W = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \mid x_1 = x_2 = x_3 \text{ and } x_4 + x_5 = 0\}$. What is the dimension of the subspace W ? **Answer: 2** **Solution:** The constraints are: $x_1 - x_2 = 0$, $x_2 - x_3 = 0$, and $x_4 + x_5 = 0$. These form 3 linearly independent equations on $\mathbb{R}^5$. Free variables: let $x_1 = t \implies x_2 = t, x_3 = t$. Let $x_4 = s \implies x_5 = -s$. Any vector in W is $t(1, 1, 1, 0, 0) + s(0, 0, 0, 1, -1)$. The subspace is spanned by 2 basis vectors, so $\dim(W) = 2$.

P1.3**[Practice | MSQ | 2 Marks]**

Which of the following subsets is/are linearly independent in $\mathbb{R}^3$?

- (A) $\{[1, 0, 0]^T, [1, 1, 0]^T, [1, 1, 1]^T\}$
- (B) $\{[1, 2, 3]^T, [2, 4, 6]^T\}$
- (C) $\{[1, 0, 2]^T, [0, 1, 3]^T, [2, 1, 7]^T\}$
- (D) $\{[2, 0, 0]^T, [0, 3, 0]^T\}$

Answer: (A, D) **Solution:** (A) Forms an upper triangular matrix with non-zero diagonal entries; determinant is $1 \neq 0$ (linearly independent). (B) The second vector is $2 \times$ the first (dependent). (C) Notice $[2, 1, 7]^T = 2[1, 0, 2]^T + [0, 1, 3]^T$ (dependent). (D) Two non-parallel orthogonal vectors in $\mathbb{R}^3$ are linearly independent.

P1.4**[Practice | NAT | 1 Mark]**

Let A be a 5×5 real matrix such that $\det(A) = 0$. What is the MAXIMUM possible number of linearly independent columns that A can have? **Answer: 4** **Solution:** Since $\det(A) = 0$, $\text{rank}(A) < 5$. The rank represents the number of linearly independent columns. Thus, the maximum rank is $5 - 1 = 4$.

P1.5**[Practice | MCQ | 2 Marks]**

Let U and W be subspaces of $\mathbb{R}^8$ with $\dim(U) = 5$ and $\dim(W) = 5$. What is the MINIMUM possible dimension of the subspace $U \cap W$?

- (A) 0

- (B) 2
- (C) 3
- (D) 5

Answer: (B) **Solution:** By Grassmann's formula: $\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$. Since $U + W$ is a subspace of $\mathbb{R}^8$, $\dim(U + W) \leq 8$. Thus, $8 \geq 5 + 5 - \dim(U \cap W) \implies 8 \geq 10 - \dim(U \cap W) \implies \dim(U \cap W) \geq 2$.

P1.6 [Practice | NAT | 2 Marks]

For what value of α are the three vectors $[1, 1, 1]^T$, $[1, 2, 3]^T$, and $[2, 3, \alpha]^T$ linearly dependent?

Answer: 4 **Solution:** Evaluate the determinant:

$$\det \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & \alpha \end{bmatrix} = 1(2\alpha - 9) - 1(\alpha - 3) + 2(3 - 2) = 2\alpha - 9 - \alpha + 3 + 2 = \alpha - 4$$

Setting $\alpha - 4 = 0 \implies \alpha = 4$.

P1.7 [Practice | MCQ | 1 Mark]

If $S = \{v_1, v_2, \dots, v_k\}$ is an orthogonal set of non-zero vectors in an inner product space V , then S is:

- (A) Always linearly independent.
- (B) Linearly dependent if $k > \dim(V)/2$.
- (C) Linearly dependent if some vector has unit norm.
- (D) A basis for V for any value of k .

Answer: (A) **Solution:** Standard linear algebra theorem: any set of mutually orthogonal non-zero vectors is strictly linearly independent.

P1.8 [Practice | MSQ | 2 Marks]

Let A be a 3×3 real symmetric matrix. Which of the following statements is/are ALWAYS true?

- (A) All eigenvalues of A are real numbers.
- (B) Eigenvectors corresponding to distinct eigenvalues of A are mutually orthogonal.
- (C) A cannot be diagonalized if it has repeated eigenvalues.
- (D) The row space and column space of A are identical.

Answer: (A, B, D) **Solution:** (A) and (B) follow from the Spectral Theorem for real symmetric matrices. (C) is false because real symmetric matrices are always orthogonally diagonalizable regardless of eigenvalue multiplicity. (D) Since $A = A^T$, its row space and column space are identical.

P1.9 [Practice | NAT | 2 Marks]

What is the dimension of the vector space of all 3×3 symmetric matrices with trace equal to zero? **Answer: 5** **Solution:** A general 3×3 symmetric matrix has $\frac{3(3+1)}{2} = 6$ independent entries (3 on the diagonal and 3 off-diagonal). The zero trace condition $a_{11} + a_{22} + a_{33} = 0$ imposes 1 linear constraint. Therefore, dimension = $6 - 1 = 5$.

P1.10 [Practice | MCQ | 1 Mark]

If vector w can be uniquely written as a linear combination of vectors in $B = \{u_1, u_2, \dots, u_m\}$, then:

- (A) The set B must be linearly independent.
- (B) w must be the zero vector.
- (C) m must equal the dimension of the ambient space.
- (D) The vectors in B must be orthogonal.

Answer: (A) **Solution:** If B were linearly dependent, there would exist non-trivial coefficients summing to zero, which could be added to any linear combination to yield non-unique representations. Thus, uniqueness implies B is linearly independent.

2 Topic 2: Clustering: K-Means (Math + Lloyd's Algorithm)

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

Consider 1D data points: $x = [2, 4, 10, 12, 16]$. Let $k = 2$ and the initial cluster centers be $c_1 = 2$ and $c_2 = 16$. After the first iteration of Lloyd's algorithm, what is the updated centroid of the first cluster? **Answer: 3.00** **Solution:** Compute distances and assign each point to the closer center (midpoint is $\frac{2+16}{2} = 9$):
For $x = 2$: closer to c_1 .
For $x = 4$: closer to c_1 .
For $x = 10, 12, 16$: closer to c_2 .
Cluster 1 contains $\{2, 4\}$. The new centroid is $\mu_1 = \frac{2+4}{2} = 3.00$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

The objective function minimized by the K -means clustering algorithm is:

- (A) Within-cluster sum of squared Euclidean distances (inertia)
- (B) Between-cluster maximum Chebyshev distance
- (C) Total Manhattan distance from the global median
- (D) Number of misclassified labels

Answer: (A) **Solution:** K -means seeks to minimize the objective $J = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|_2^2$, which is the within-cluster sum of squared errors (WCSS or inertia).

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about the K -means clustering algorithm is/are TRUE?

- (A) Lloyd's algorithm is guaranteed to converge in a finite number of iterations.
- (B) Lloyd's algorithm is guaranteed to converge to the global minimum of the objective function.
- (C) K -means is sensitive to outliers and initial centroid positions.
- (D) K -means performs best when clusters are spherical and have similar sizes.

Answer: (A, C, D) **Solution:** (A) There are a finite number of possible partitionings (K^n), and each step strictly decreases or maintains the objective, ensuring finite convergence. (B) is false because it commonly gets trapped in local minima. (C) Outliers heavily distort the mean, and bad initialization causes sub-optimal local minima. (D) Euclidean distance induces spherical Voronoi decision cells.

Q2.4

[GATE CS 2021 | NAT | 2 Marks]

Consider four 2D points: $A(1, 1)$, $B(1, 3)$, $C(5, 1)$, and $D(5, 3)$. If $k = 2$ and the two clusters obtained are $C_1 = \{A, B\}$ and $C_2 = \{C, D\}$, what is the total within-cluster sum of squares (WCSS)?

Answer: 4.00 **Solution:** Centroid of C_1 : $\mu_1 = (\frac{1+1}{2}, \frac{1+3}{2}) = (1, 2)$.

Sum of squares for C_1 : $\|(1, 1) - (1, 2)\|^2 + \|(1, 3) - (1, 2)\|^2 = 1^2 + 1^2 = 2$.

Centroid of C_2 : $\mu_2 = (\frac{5+5}{2}, \frac{1+3}{2}) = (5, 2)$.

Sum of squares for C_2 : $\|(5, 1) - (5, 2)\|^2 + \|(5, 3) - (5, 2)\|^2 = 1^2 + 1^2 = 2$.

Total WCSS = $2 + 2 = 4.00$.

Q2.5

[GATE DA 2025 | MCQ | 1 Mark]

How does the K -means++ algorithm improve upon standard random initialization in K -means?

- (A) It chooses initial cluster centers with probability proportional to their squared distance from the closest existing center.

- (B) It runs agglomerative clustering first to select centers.
- (C) It places all initial centers at the coordinates of the first K data points.
- (D) It sets all cluster centers at the global sample mean.

Answer: (A) Solution: K -means++ initializes the first center uniformly at random and then selects each subsequent center from remaining points with probability $P(x) = \frac{D(x)^2}{\sum D(x')^2}$, where $D(x)$ is the distance to the nearest already chosen center, spreading them apart.

Q2.6 [GATE CS 2022 | MCQ | 2 Marks]

What is the worst-case time complexity of one single iteration of Lloyd's algorithm for n data points in d dimensions with k clusters?

- (A) $O(n \cdot k \cdot d)$
- (B) $O(n^2 \cdot d)$
- (C) $O(k^2 \cdot d)$
- (D) $O(n \cdot \log k)$

Answer: (A) Solution: In the assignment step, for each of the n points, the Euclidean distance to all k centroids is computed, taking $O(k \cdot d)$ per point ($O(nkd)$ total). In the update step, recomputing centroids takes $O(nd)$. Thus, one iteration takes $O(nkd)$ time.

Q2.7 [GATE DA 2026 | NAT | 2 Marks]

Let a dataset consist of 6 points along the real line: $\{1, 2, 3, 7, 8, 9\}$. If $k = 2$ and the true natural clusters are $C_1 = \{1, 2, 3\}$ and $C_2 = \{7, 8, 9\}$, what is the total WCSS? **Answer: 4.00** **Solution:**

Centroid of C_1 : $\mu_1 = \frac{1+2+3}{3} = 2$.

$SS_1 = (1-2)^2 + (2-2)^2 + (3-2)^2 = 1 + 0 + 1 = 2$.

Centroid of C_2 : $\mu_2 = \frac{7+8+9}{3} = 8$.

$SS_2 = (7-8)^2 + (8-8)^2 + (9-8)^2 = 1 + 0 + 1 = 2$.

Total WCSS = $2 + 2 = 4.00$.

Q2.8 [GATE CS 2020 | MCQ | 1 Mark]

If the number of clusters k in K -means is set equal to the number of data points n ($k = n$), the resulting value of the within-cluster sum of squares (WCSS) objective is:

- (A) 0
- (B) ∞
- (C) Total variance of the dataset
- (D) Undefined

Answer: (A) Solution: When $k = n$, each data point is its own cluster center ($c_i = x_i$). The distance from each point to its assigned centroid is zero, so $WCSS = \sum_{i=1}^n \|x_i - x_i\|^2 = 0$.

Q2.9 [GATE DA 2024 | MSQ | 2 Marks]

In the context of evaluating K -means clustering, which of the following statements is/are TRUE?

- (A) The Elbow method looks for the point where the rate of decrease in WCSS slows down significantly.
- (B) The Silhouette coefficient of a point is bounded between -1 and $+1$.
- (C) A Silhouette coefficient close to $+1$ indicates that the point is well-matched to its own cluster and poorly matched to neighboring clusters.

(D) The value of WCSS always increases as k increases.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are fundamental cluster validation facts. (D) is false: increasing k provides more centroids to fit points, so WCSS monotonically decreases (or stays flat), never increases.

Q2.10

[GATE CS 2019 | NAT | 1 Mark]

A cluster consists of three 2D points: $(2, 3)$, $(4, 7)$, and $(6, 5)$. What is the y -coordinate of the cluster centroid? **Answer: 5.00** **Solution:** Centroid y -coordinate is $\bar{y} = \frac{3+7+5}{3} = \frac{15}{3} = 5.00$.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

Mathematically, why does setting the cluster centroid to the sample mean of its points minimize the within-cluster sum of squares?

- (A) Because the mean minimizes the sum of squared Euclidean deviations $\sum \|x_i - \mu\|^2$.
- (B) Because the mean minimizes the sum of absolute deviations $\sum |x_i - \mu|$.
- (C) Because the mean maximizes the between-cluster covariance.
- (D) Because the mean eliminates outliers.

Answer: (A) **Solution:** Differentiating $\sum_{i=1}^m \|x_i - \mu\|^2$ with respect to μ and setting the derivative to zero gives $-2 \sum_{i=1}^m (x_i - \mu) = 0 \implies \mu = \frac{1}{m} \sum x_i$, which is the arithmetic mean.

P2.2

[Practice | NAT | 2 Marks]

Points on a line are: $x = [1, 3, 5, 15, 17, 19]$. Two initial centers are chosen as $c_1 = 4$ and $c_2 = 12$. What is the updated center c_2 after the first assignment step? **Answer: 17.00** **Solution:** Decision boundary between $c_1 = 4$ and $c_2 = 12$ is $\frac{4+12}{2} = 8$. Points ≤ 8 belong to C_1 : $\{1, 3, 5\}$. Points > 8 belong to C_2 : $\{15, 17, 19\}$. The new centroid of C_2 is $\frac{15+17+19}{3} = \frac{51}{3} = 17.00$.

P2.3

[Practice | MSQ | 2 Marks]

Which of the following modifications can be used when dealing with datasets that have non-spherical (e.g., concentric circles or crescent) cluster geometries?

- (A) Kernel K -means
- (B) Standard Lloyd's algorithm with Euclidean distance
- (C) Spectral Clustering
- (D) DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Answer: (A, C, D) **Solution:** Standard K -means with Euclidean distance is restricted to convex spherical clusters. Kernel K -means (A), Spectral clustering (C), and DBSCAN (D) successfully partition arbitrary non-convex geometries.

P2.4

[Practice | NAT | 2 Marks]

For a single data point i , the mean intra-cluster distance is $a(i) = 2.0$ and the mean nearest-cluster distance is $b(i) = 6.0$. What is the Silhouette coefficient $s(i)$? **Answer: 0.67** **Solution:** The Silhouette coefficient formula is $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} = \frac{6.0 - 2.0}{\max(2.0, 6.0)} = \frac{4.0}{6.0} \approx 0.67$.

P2.5

[Practice | MCQ | 1 Mark]

If K -medoids (PAM algorithm) is used instead of K -means, the cluster center is restricted to:

- (A) Any arbitrary point in the continuous feature space.

- (B) One of the actual observations belonging to the cluster.
- (C) The geometric median computed via gradient descent.
- (D) The origin of the coordinate system.

Answer: (B) **Solution:** K -medoids restricts each cluster center to be an actual exemplar data point from the dataset, which provides robustness against extreme outliers.

P2.6 [Practice | NAT | 2 Marks]

Consider 3 points in 1D: $x_1 = 0, x_2 = 4, x_3 = 8$. If all three points are grouped into a single cluster ($k = 1$), what is the value of WCSS? **Answer: 32.00** **Solution:** Centroid $\mu = \frac{0+4+8}{3} = 4$.
 $WCSS = (0 - 4)^2 + (4 - 4)^2 + (8 - 4)^2 = (-4)^2 + 0^2 + 4^2 = 16 + 0 + 16 = 32.00$.

P2.7 [Practice | MCQ | 2 Marks]

What is the guaranteed theoretical approximation ratio of the K -means++ seeding algorithm relative to the optimal K -means clustering objective?

- (A) $O(1)$
- (B) $O(\log K)$
- (C) $O(K)$
- (D) $O(\sqrt{n})$

Answer: (B) **Solution:** Arthur and Vassilvitskii (2007) proved that K -means++ achieves an expected approximation ratio of $O(\log K)$, meaning $\mathbb{E}[\phi] \leq 8(\ln K + 2)\phi_{OPT}$.

P2.8 [Practice | MSQ | 2 Marks]

During the execution of Lloyd's algorithm for K -means:

- (A) The objective function (WCSS) monotonically decreases or stays constant at each iteration.
- (B) The algorithm can oscillate indefinitely between two non-optimal cluster configurations.
- (C) A cluster can become empty if no points are assigned to its centroid.
- (D) Feature scaling (e.g., standardization) affects the resulting clusters.

Answer: (A, C, D) **Solution:** (A) Both assignment and update steps minimize the same objective function, ensuring monotonic descent. (B) is false; monotonic descent over a finite set of partitions prevents cycling. (C) Empty clusters can occur. (D) Because Euclidean distance is scale-dependent, unscaled features with large variances dominate the distance calculations.

P2.9 [Practice | NAT | 1 Mark]

A dataset has 50 points. If K -means is run with $k = 1$, the total sum of squares about the global mean is 500. When $k = 4$, the between-cluster sum of squares (BCSS) is 380. What is the value of WCSS? **Answer: 120** **Solution:** By the total variance decomposition identity: $TSS = WCSS + BCSS$. Thus, $WCSS = TSS - BCSS = 500 - 380 = 120$.

P2.10 [Practice | MCQ | 1 Mark]

Which of the following stop conditions is NOT commonly used to terminate Lloyd's algorithm?

- (A) Centroid coordinates change by less than a tolerance threshold ϵ .
- (B) Cluster assignments of all data points remain unchanged.
- (C) A maximum predetermined number of iterations is reached.
- (D) The error rate on a separate test validation set reaches zero.

Answer: (D) **Solution:** K -means is an unsupervised clustering algorithm with no ground-truth class labels, so test validation error rates are not applicable.

3 Topic 3: Transactions & ACID Properties

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 1 Mark]

Which of the following ACID properties ensures that all operations of a database transaction are completed successfully, or in the event of a failure, none of them take effect?

- (A) Atomicity
- (B) Consistency
- (C) Isolation
- (D) Durability

Answer: (A) Solution: Atomicity embodies the “all-or-nothing” property: either the transaction completes entirely and its updates are committed, or it is aborted and all intermediate changes are rolled back.

Q3.2

[GATE CS 2023 | MCQ | 2 Marks]

Consider a concurrent execution of two transactions T_1 and T_2 :

$$S : R_1(A); W_1(A); R_2(A); W_2(A); R_1(B); W_1(B); \text{Commit}_1; \text{Commit}_2$$

The schedule S is:

- (A) Conflict serializable
- (B) Not conflict serializable
- (C) View serializable but not conflict serializable
- (D) A strict schedule

Answer: (A) Solution: Find conflicting operations (same data item, different transactions, at least one write):

Conflicting pairs on A : $(R_1(A), W_2(A)) \implies T_1 \rightarrow T_2; (W_1(A), R_2(A)) \implies T_1 \rightarrow T_2; (W_1(A), W_2(A)) \implies T_1 \rightarrow T_2$.

There are no conflicts where T_2 precedes T_1 . The precedence (serialization) graph has only edge $T_1 \rightarrow T_2$, which is acyclic. Hence, S is conflict serializable.

Q3.3

[GATE DA 2024 | MCQ | 1 Mark]

The Write-Ahead Logging (WAL) protocol used by database recovery systems enforces which of the following rules?

- (A) The log record for an update must be flushed to stable storage before the corresponding modified data page is written to disk.
- (B) All data pages must be written to disk before writing the log record.
- (C) All transaction commits must be delayed until the database checkpoint finishes.
- (D) Log records are kept only in volatile main memory.

Answer: (A) Solution: The WAL rule mandates that log records describing a change must reach stable disk storage before the corresponding dirty data page is written to disk, ensuring Atomicity and Durability.

Q3.4

[GATE CS 2022 | NAT | 2 Marks]

Consider the schedule S with transactions T_1, T_2, T_3 :

$S : R_1(X); R_2(Y); R_3(X); W_1(Y); W_2(Z); W_3(X); \text{Commit}_1; \text{Commit}_2; \text{Commit}_3$

How many directed edges exist in the precedence graph of S ? **Answer: 2** **Solution:** Check conflicting operations between transactions:

1. On X : $R_1(X)$ followed by $W_3(X) \implies T_1 \rightarrow T_3$. Also $R_3(X)$ and $W_3(X)$ are same transaction.
 2. On Y : $R_2(Y)$ followed by $W_1(Y) \implies T_2 \rightarrow T_1$.
 3. On Z : only $W_2(Z)$, no other transaction accesses Z .
- Edges: $T_2 \rightarrow T_1$ and $T_1 \rightarrow T_3$. Total directed edges = 2.

Q3.5

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following database anomalies can occur when transactions execute concurrently without adequate isolation?

- (A) Dirty Read (Reading uncommitted data)
- (B) Non-repeatable Read (Fuzzy Read)
- (C) Phantom Read
- (D) Lost Update

Answer: (A, B, C, D) **Solution:** All four are classic concurrency anomalies identified in ANSI SQL-92 standards resulting from insufficient isolation levels (Read Uncommitted, Read Committed, Repeatable Read).

Q3.6

[GATE CS 2021 | MCQ | 2 Marks]

A schedule is considered "recoverable" if:

- (A) For every transaction T_j that reads a data item written by T_i , the commit of T_i precedes the commit of T_j .
- (B) No transaction reads data written by an uncommitted transaction.
- (C) It is conflict serializable.
- (D) The precedence graph contains no directed cycles.

Answer: (A) **Solution:** By definition, a schedule is recoverable if whenever transaction T_j reads a value updated by T_i ($T_i \rightarrow T_j$ dirty read), T_i commits before T_j commits.

Q3.7

[GATE CS 2020 | NAT | 1 Mark]

Under the strict two-phase locking (Strict 2PL) protocol, all exclusive locks acquired by a transaction are held until:

- (A) The end of the transaction (Commit or Abort)
- (B) The end of the growing phase
- (C) The first read operation
- (D) The beginning of the shrinking phase

Answer: (A) **Solution:** Strict 2PL requires all exclusive (X) locks to be held until the transaction finishes (commits or aborts), which prevents dirty reads and ensures cascading-abort-free execution.

Q3.8

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following statements about the ACID properties is/are TRUE?

- (A) Consistency ensures that a transaction takes the database from one valid state satisfying all integrity constraints to another valid state.
- (B) Durability ensures that committed updates survive system crashes and power failures.
- (C) The Recovery Manager is primarily responsible for ensuring Atomicity and Durability.
- (D) The Concurrency Control Manager is primarily responsible for ensuring Isolation.

Answer: (A, B, C, D) **Solution:** All four are fundamental database definitions. Consistency maintains integrity invariants. Durability guarantees persistence via write-ahead logging and checkpoints. Recovery manages A and D; concurrency control manages I.

Q3.9

[GATE CS 2019 | MCQ | 2 Marks]

Consider the following partial schedule:

$S : W_1(X); R_2(X); W_2(Y); \text{Abort}_1$

If T_2 is allowed to commit before T_1 aborts, what type of anomaly occurs?

- (A) Irrecoverable schedule
- (B) Phantom read
- (C) Lost update
- (D) Write-write conflict

Answer: (A) **Solution:** T_2 reads uncommitted data written by T_1 . If T_2 commits and then T_1 aborts, the database cannot rollback T_2 's committed effects, leading to an irrecoverable schedule.

Q3.10

[GATE CS 2018 | NAT | 1 Mark]

How many serial schedules are possible for a set of 4 concurrent transactions $\{T_1, T_2, T_3, T_4\}$?

Answer: 24 **Solution:** Any serial execution corresponds to a permutation of the 4 transactions. The number of permutations is $4! = 4 \times 3 \times 2 \times 1 = 24$.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

In SQL-92, which isolation level prevents Dirty Reads and Non-repeatable Reads, but may still permit Phantom Reads?

- (A) Read Uncommitted
- (B) Read Committed
- (C) Repeatable Read
- (D) Serializable

Answer: (C) **Solution:** Repeatable Read locks rows read by the transaction, preventing dirty and non-repeatable reads. However, range query inserts can still insert new matching rows, allowing Phantom Reads.

P3.2

[Practice | NAT | 2 Marks]

Consider a schedule with 3 transactions where the precedence graph has vertices $\{T_1, T_2, T_3\}$ and directed edges $T_1 \rightarrow T_2$ and $T_2 \rightarrow T_3$. How many topological sortings (equivalent serial orders) exist? **Answer: 1** **Solution:** The edges require T_1 before T_2 , and T_2 before T_3 . The only valid linear ordering is $T_1 \rightarrow T_2 \rightarrow T_3$. Thus, there is exactly 1 equivalent serial order.

P3.3

[Practice | MSQ | 2 Marks]

Which of the following schedules is/are conflict serializable?

- (A) $S_1 : R_1(X); W_2(X); W_1(X);$
- (B) $S_2 : R_1(X); W_1(X); R_2(X); W_2(X);$
- (C) $S_3 : R_1(X); R_2(Y); W_1(Y); W_2(X);$
- (D) $S_4 : R_1(X); W_1(Y); R_2(Y); W_2(Z);$

Answer: (B, D) Solution: In $S_1: R_1(X) \rightarrow W_2(X) \Rightarrow T_1 \rightarrow T_2$, but $W_2(X) \rightarrow W_1(X) \Rightarrow T_2 \rightarrow T_1$ (cycle: non-serializable).

In S_2 : Conflicts give only $T_1 \rightarrow T_2$ (acyclic: serializable).

In $S_3: R_2(Y) \rightarrow W_1(Y) \Rightarrow T_2 \rightarrow T_1$, and $R_1(X) \rightarrow W_2(X) \Rightarrow T_1 \rightarrow T_2$ (cycle: non-serializable).

In S_4 : Conflicting pair on Y gives $W_1(Y) \rightarrow R_2(Y) \Rightarrow T_1 \rightarrow T_2$ (acyclic: serializable).

P3.4

[Practice | MCQ | 1 Mark]

A schedule is "cascadeless" (avoids cascading aborts) if:

- (A) Every transaction reads only values written by committed transactions.
- (B) All transactions acquire shared locks before exclusive locks.
- (C) No transaction executes any write operation.
- (D) The precedence graph is a tree.

Answer: (A) Solution: A schedule avoids cascading aborts if transactions read only data items that have already been committed by the writer, eliminating the need to rollback dependent readers if a writer aborts.

P3.5

[Practice | NAT | 2 Marks]

In timestamp-based concurrency control, let transaction T have timestamp $TS(T) = 50$. Suppose T issues $Write(Q)$. If the read timestamp of data item Q is $R-TS(Q) = 60$, will the write operation be accepted? Enter 1 for Yes, 0 for No. **Answer: 0 Solution:** Under the standard Thomas/Timestamp Ordering protocol, if $TS(T) < R-TS(Q)$, a younger transaction with timestamp 60 has already read the value of Q , assuming T 's write had already occurred. Thus, T 's write is rejected and T is aborted and restarted.

P3.6

[Practice | MCQ | 2 Marks]

What is the relationship between the sets of Conflict Serializable (CSR) and View Serializable (VSR) schedules?

- (A) $CSR \subset VSR$ (CSR is a strict subset of VSR)
- (B) $VSR \subset CSR$
- (C) $CSR = VSR$
- (D) $CSR \cap VSR = \emptyset$

Answer: (A) Solution: Every conflict serializable schedule is also view serializable, but there exist view serializable schedules (relying on blind writes) that are not conflict serializable. Thus, $CSR \subset VSR$.

P3.7

[Practice | MSQ | 2 Marks]

Which of the following statements about deadlock in transaction management is/are TRUE?

- (A) Standard Two-Phase Locking (2PL) does not guarantee freedom from deadlocks.
- (B) Deadlocks can be prevented using the Wait-Die or Wound-Wait timestamp schemes.
- (C) A directed cycle in the Wait-For Graph (WFG) indicates the existence of a deadlock.

(D) Strict 2PL prevents deadlocks completely.

Answer: (A, B, C) Solution: (A) Basic 2PL and Strict 2PL can encounter deadlocks (e.g., T_1 holds lock on A and requests B ; T_2 holds lock on B and requests A). (B) Timestamp preemption prevents hold-and-wait cycles. (C) WFG cycles are the necessary and sufficient condition for deadlock in single-unit resources. (D) is false.

P3.8

[Practice | NAT | 1 Mark]

Consider the schedule $S : R_1(X); W_2(X); W_1(X); \text{Commit}_1; \text{Commit}_2$. Is S view serializable? Enter 1 for Yes, 0 for No. **Answer: 0 Solution:** Check serial orders: If T_1, T_2 , final write on X is by T_2 , but in S it is by T_1 . If T_2, T_1, T_1 reads the value written by T_2 , but in S, T_1 reads the initial state. Thus, S is not view equivalent to any serial schedule (0).

P3.9

[Practice | MCQ | 1 Mark]

During database recovery using the ARIES algorithm, the three phases executed in order are:

- (A) Analysis, Redo, Undo
- (B) Undo, Redo, Analysis
- (C) Redo, Analysis, Undo
- (D) Checkpoint, Rollback, Commit

Answer: (A) Solution: The classic ARIES recovery protocol executes: (1) Analysis phase (scans log forward from checkpoint), (2) Redo phase (repeats history forward), and (3) Undo phase (rolls back active uncommitted transactions backward).

P3.10

[Practice | NAT | 2 Marks]

How many conflicting pairs of operations exist in the following schedule?

$S : R_1(A); W_2(A); R_3(A); W_1(A)$

Answer: 4 Solution: Examine pairs on same item A belonging to different transactions where at least one is a write:

1. $(R_1(A), W_2(A))$
2. $(R_1(A), W_1(A))$ is same transaction $\implies$ not conflicting across transactions.
3. $(W_2(A), R_3(A))$
4. $(W_2(A), W_1(A))$
5. $(R_3(A), W_1(A))$

Total pairs = 4.

4 Topic 4: Data Interpretation – Tables & Bar Graphs

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

The table below shows the production (in metric tons) of rice by four states over two consecutive years:

State	Year 2022	Year 2023
P	120	144
Q	150	165
R	80	100
S	200	220

Which state experienced the highest percentage growth in rice production from 2022 to 2023?

- (A) State P
- (B) State Q
- (C) State R
- (D) State S

Answer: (C) **Solution:** Percentage growths:

State P : $\frac{144-120}{120} \times 100\% = \frac{24}{120} \times 100\% = 20\%$.

State Q : $\frac{165-150}{150} \times 100\% = \frac{15}{150} \times 100\% = 10\%$.

State R : $\frac{100-80}{80} \times 100\% = \frac{20}{80} \times 100\% = 25\%$.

State S : $\frac{220-200}{200} \times 100\% = \frac{20}{200} \times 100\% = 10\%$.

State R recorded the highest growth at 25%.

Q4.2

[GATE GA 2023 | NAT | 2 Marks]

The bar graph shows the number of units of an electronic item sold by three branches B_1, B_2, B_3 over three years (in thousands):

Branch	2020	2021	2022
B_1	40	50	60
B_2	30	45	55
B_3	50	55	65

What is the average number of units sold (in thousands) by Branch B_2 over the three years? **Answer: 43.33**

Solution: Sum of sales for $B_2 = 30 + 45 + 55 = 130$. Average $= \frac{130}{3} \approx 43.33$ thousand.

Q4.3

[GATE GA 2022 | MCQ | 2 Marks]

The table shows the percentage of passed students and total students appeared in an examination from four institutes:

Institute	Appeared	Pass Percentage (%)
A	400	75%
B	500	60%
C	300	80%
D	600	50%

What is the overall pass percentage across all four institutes combined?

- (A) 62.5%

- (B) 63.89%
- (C) 65.0%
- (D) 66.25%

Answer: (B) **Solution:** Calculate passed students:

A: $0.75 \times 400 = 300$.

B: $0.60 \times 500 = 300$.

C: $0.80 \times 300 = 240$.

D: $0.50 \times 600 = 300$.

Total passed = $300 + 300 + 240 + 300 = 1140$.

Total appeared = $400 + 500 + 300 + 600 = 1800$.

Overall pass percentage = $\frac{1140}{1800} \times 100\% = \frac{114}{180} \times 100\% = \frac{19}{30} \times 100\% \approx 63.89\%$.

Q4.4

[GATE GA 2021 | NAT | 1 Mark]

The revenues of a startup in four quarters (Q1, Q2, Q3, Q4) are Rs. 10 lakhs, Rs. 15 lakhs, Rs. 25 lakhs, and Rs. 30 lakhs, respectively. What is the percentage increase in revenue from Q1 to Q4?

Answer: 200 **Solution:** Increase = $30 - 10 = 20$ lakhs. Percentage increase = $\frac{20}{10} \times 100\% = 200\%$.

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

The bar chart represents expenditure on raw materials and salaries (in lakhs):

Year	Raw Material	Salaries
2021	45	30
2022	60	40
2023	80	50

The ratio of total expenditure on raw materials to total expenditure on salaries across all three years is:

- (A) 37 : 24
- (B) 185 : 120
- (C) 31 : 20
- (D) 5 : 3

Answer: (A) **Solution:** Total raw materials = $45 + 60 + 80 = 185$. Total salaries = $30 + 40 + 50 = 120$. Ratio = $\frac{185}{120} = \frac{37}{24}$.

Q4.6

[GATE GA 2020 | NAT | 2 Marks]

The table presents employee counts in different departments across two offices:

Department	Office 1	Office 2
HR	25	15
Tech	120	180
Marketing	40	60
Finance	15	25

What percentage of the total workforce across both offices is employed in the Tech department? (Round off to one decimal place) **Answer: 62.5** **Solution:** Total Tech employees = $120 + 180 = 300$.

Total workforce = $(25 + 15) + (120 + 180) + (40 + 60) + (15 + 25) = 40 + 300 + 100 + 40 = 480$.

Percentage = $\frac{300}{480} \times 100\% = \frac{5}{8} \times 100\% = 62.5\%$.

Q4.7

[GATE GA 2025 | MCQ | 1 Mark]

The import and export data of a commodity (in million dollars) is shown below:

Year	Imports	Exports
2021	120	140
2022	150	135
2023	160	190
2024	200	210

In which year was the trade deficit (Imports – Exports) positive and highest?

- (A) 2021
- (B) 2022
- (C) 2023
- (D) 2024

Answer: (B) Solution: Trade deficit = Imports – Exports.

2021: $120 - 140 = -20$ (Surplus).

2022: $150 - 135 = +15$ (Deficit of 15).

2023: $160 - 190 = -30$ (Surplus).

2024: $200 - 210 = -10$ (Surplus).

Only 2022 has a positive trade deficit.

Q4.8

[GATE GA 2023 | NAT | 2 Marks]

The table shows monthly power generation (in GWh) from Solar and Wind sources:

Month	Solar	Wind
Jan	450	350
Feb	500	400
Mar	550	450

By what percentage does total power generation in March exceed total power generation in January? **Answer: 25** **Solution:** Total generation in Jan = $450 + 350 = 800$ GWh. Total in March = $550 + 450 = 1000$ GWh. Percentage increase = $\frac{1000-800}{800} \times 100\% = \frac{200}{800} \times 100\% = 25\%$.

Q4.9

[GATE GA 2026 | MCQ | 1 Mark]

A company's profits (in Rs. Crores) over four years are: 2021: 15, 2022: 25, 2023: 20, 2024: 30. Which year registered the maximum percentage increase in profit compared to the preceding year?

- (A) 2022
- (B) 2023
- (C) 2024
- (D) Both 2022 and 2024

Answer: (A) Solution: Percentage changes:

2021 to 2022: $\frac{25-15}{15} \times 100\% = \frac{10}{15} \times 100\% = 66.67\%$.

2022 to 2023: Negative (decline).

2023 to 2024: $\frac{30-20}{20} \times 100\% = \frac{10}{20} \times 100\% = 50\%$.

The maximum percentage increase occurred in 2022 (66.67%).

Q4.10

[GATE GA 2019 | NAT | 1 Mark]

A car manufacturer produces Sedan and SUV models. In 2022, 60% of total cars produced were Sedans. If 12,000 Sedans were produced, what was the number of SUVs produced? **Answer:**

8000 **Solution:** Let total cars be N . $0.60N = 12000 \implies N = 20000$. Number of SUVs = $0.40N = 0.40 \times 20000 = 8000$.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | MCQ | 1 Mark]

The table presents the monthly production cost and sales revenue of an enterprise:

Month	Cost (Rs. Lakhs)	Revenue (Rs. Lakhs)
Jan	40	50
Feb	50	65
Mar	60	75
Apr	80	104

Which month had the highest profit margin percentage ($\frac{\text{Profit}}{\text{Cost}} \times 100\%$)?

- (A) Jan
- (B) Feb
- (C) Mar
- (D) Apr

Answer: (B) **Solution:** Profit margins:

$$\text{Jan: } \frac{50-40}{40} \times 100\% = \frac{10}{40} = 25\%.$$

$$\text{Feb: } \frac{65-50}{50} \times 100\% = \frac{15}{50} = 30\%.$$

$$\text{Mar: } \frac{75-60}{60} \times 100\% = \frac{15}{60} = 25\%.$$

$$\text{Apr: } \frac{104-80}{80} \times 100\% = \frac{24}{80} = 30\%.$$

Both Feb and Apr tie at 30%; choosing Feb.

P4.2

[Practice | NAT | 2 Marks]

The bar graph shows the population (in millions) of two cities X and Y over three decades:

City	2000	2010	2020
X	4.0	5.0	6.5
Y	3.0	4.2	6.0

What is the difference in population growth rate (percentage increase from 2000 to 2020) between City Y and City X ? **Answer: 37.5** **Solution:** Growth rate of $X = \frac{6.5-4.0}{4.0} \times 100\% = \frac{2.5}{4.0} \times 100\% = 62.5\%$.

$$\text{Growth rate of } Y = \frac{6.0-3.0}{3.0} \times 100\% = 100\%.$$

$$\text{Difference} = 100\% - 62.5\% = 37.5\%.$$

P4.3

[Practice | MSQ | 2 Marks]

A software firm evaluates bugs reported and fixed across 4 modules:

Module	Reported	Fixed
M_1	120	96
M_2	200	170
M_3	150	105
M_4	80	68

Which of the following statements is/are TRUE?

- (A) Module M_2 has the highest absolute number of fixed bugs.
- (B) Module M_4 has a higher fix rate ($\frac{\text{Fixed}}{\text{Reported}}$) than Module M_1 .
- (C) Module M_3 has the lowest bug fix rate.

(D) The overall fix rate across all modules exceeds 80%.

Answer: (A, B, C, D) **Solution:** Fix rates:

$$M_1: \frac{96}{120} = 80\%.$$

$$M_2: \frac{170}{200} = 85\% \text{ (highest count 170).}$$

$$M_3: \frac{105}{150} = 70\% \text{ (lowest rate).}$$

$$M_4: \frac{68}{80} = 85\% > 80\%.$$

Total fixed = $96 + 170 + 105 + 68 = 439$. Total reported = $120 + 200 + 150 + 80 = 550$. Overall fix rate = $\frac{439}{550} \times 100\% \approx 79.82\%$? Wait, $439/550 = 0.79818 < 80\%$. So (D) is false. Statements (A, B, C) are TRUE.

P4.4

[Practice | NAT | 2 Marks]

The table presents student enrollment across four years in a college:

Year	1st Year	2nd Year	3rd Year	4th Year
Count	420	380	350	310

What is the average number of students enrolled per year? **Answer: 365** **Solution:** Total enrollment = $420 + 380 + 350 + 310 = 1460$. Average = $\frac{1460}{4} = 365$.

P4.5

[Practice | MCQ | 1 Mark]

The sales of three models of cars A, B, C are shown below:

Year	Model A	Model B	Model C
2022	150	120	80
2023	180	150	120

Which model showed the highest percentage increase in sales in 2023 over 2022?

- (A) Model A
- (B) Model B
- (C) Model C
- (D) All three models had equal percentage growth

Answer: (C) **Solution:** Model A: $\frac{180-150}{150} = \frac{30}{150} = 20\%$.

Model B: $\frac{150-120}{120} = \frac{30}{120} = 25\%$.

Model C: $\frac{120-80}{80} = \frac{40}{80} = 50\%$.

Model C had the highest growth (50%).

P4.6

[Practice | NAT | 1 Mark]

In an office of 120 employees, a survey recorded daily coffee consumption: 30 drink 0 cups, 50 drink 1 cup, 30 drink 2 cups, and 10 drink 3 cups. What is the total number of cups of coffee consumed daily? **Answer: 140** **Solution:** Total cups = $(30 \times 0) + (50 \times 1) + (30 \times 2) + (10 \times 3) = 0 + 50 + 60 + 30 = 140$.

P4.7

[Practice | MCQ | 2 Marks]

The table gives the speed and travel time for four delivery vehicles:

Vehicle	Speed (km/h)	Time (hours)
V_1	40	3.5
V_2	60	2.5
V_3	50	3.0
V_4	75	2.0

Which vehicle covered the maximum total distance?

- (A) V_1
- (B) V_2
- (C) V_3
- (D) Both V_2 and V_4

Answer: (D) **Solution:** Distances ($D = S \times T$):

$$V_1: 40 \times 3.5 = 140 \text{ km.}$$

$$V_2: 60 \times 2.5 = 150 \text{ km.}$$

$$V_3: 50 \times 3.0 = 150 \text{ km.}$$

$$V_4: 75 \times 2.0 = 150 \text{ km.}$$

Wait, V_2, V_3, V_4 all cover 150 km. Thus V_2, V_3, V_4 tie at the maximum distance of 150 km.

P4.8

[Practice | NAT | 2 Marks]

The bar graph shows quarterly expenditures on cloud infrastructure (in thousands of dollars): Q1: 45, Q2: 55, Q3: 65, Q4: 75. If cloud spend increases by 20% in Q1 of the next year compared to Q4, what will be the Q1 spend of the next year? **Answer: 90** **Solution:** Next year Q1 spend = $75 \times 1.20 = 90$ thousand dollars.

P4.9

[Practice | MSQ | 2 Marks]

The table presents passing stats for male and female candidates:

Category	Appeared (M)	Passed (M)	Appeared (F)	Passed (F)
Urban	200	160	150	135
Rural	300	180	250	175

Which of the following statements is/are TRUE?

- (A) In the Urban category, female candidates have a higher pass percentage than male candidates.
- (B) The overall pass percentage for rural candidates is 70%.
- (C) More male candidates passed in total than female candidates.
- (D) The total number of candidates who passed is 650.

Answer: (A, C, D) **Solution:** (A) Urban Male: $160/200 = 80\%$; Urban Female: $135/150 = 90\%$ (True).

(B) Rural pass rate: $\frac{180+175}{300+250} = \frac{355}{550} \approx 64.55\% \neq 70\%$ (False).

(C) Total passed males = $160 + 180 = 340$; Total passed females = $135 + 175 = 310$. $340 > 310$ (True).

(D) Total passed = $340 + 310 = 650$ (True).

P4.10

[Practice | NAT | 1 Mark]

In a university department, the ratio of graduate to undergraduate students is 3 : 7. If there are 210 graduate students, what is the total number of students in the department? **Answer: 700**

Solution: Let 1 unit be k . $3k = 210 \implies k = 70$. Total students = $3k + 7k = 10k = 10 \times 70 = 700$.